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Marketplace Expansion Through Marquee Seller Adoption: Externalities and Reputation Implications

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Abstract. In the race to establish themselves, many early-stage online marketplaces choose to accelerate their growth by adding marquee (established brand name) sellers. We study the implications of marquee seller entry on smaller, unbranded sellers in a marketplace when both unbranded sellers and marquee sellers can vary *vertically* across reputation (referred to as sellers' *quality*). Although recent literature has shown that higher-quality unbranded sellers fare better than their lower-quality peers, we posit that this may not hold for entrants of any quality. To this end, we collaborate with an online business-to-business platform and exploit the entry of two marquee sellers of vastly differing quality. Using a difference-in-differences framework, we causally identify the effect. We find that, although higher-quality unbranded seller revenues increase relative to low-quality unbranded sellers when the entrant is of superior quality (consistent with the literature), the effect is *reversed* when the entrant is of inferior quality. Further, unbranded sellers change their supply quantities such that the platform's average supply quality shifts in the direction of entrant quality. Using a stylized theoretical model, we identify two mechanisms that drive our findings: (i) new buyers brought in by the entrant disproportionately favor unbranded sellers who are quality neighbors to the entrant and (ii) the unbranded seller's ability to adjust their supply quantities. Most notably, the choice of marquee sellers, examined through the lens of their externality on unbranded sellers, can foster or undermine the platform's long-term growth objectives.

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Keywords: online marketplaces • seller reputation • marquee adoption • expansion strategy • empirical operations management • difference-in-difference-in-differences

1. Introduction

In 2007, the idea of Airbnb was born when its founders rented out three airbeds in their San Francisco apartment. Today, Airbnb boasts millions of rental listings, with hosts varying in reputation (quality) and price: a far cry from the initial offering portfolio. Although the ability to grow and expand offerings is imperative for many businesses to succeed, it is much more so for such online platforms where it can be particularly challenging to keep and grow the buyer base due to expanding options and low search costs. Online platforms offer the opportunity of ease-of-connection and positive network externalities to both buyers/users and sellers; the larger the offerings, the higher the value to potential buyers/users. Growth for these platforms is therefore not only key to more growth but also to profitability and long-term survival.

Although growth on one side of the market can drive growth on the other side and a positive cycle of growth

can emerge, it is often challenging to initiate a positive cycle of growth (i.e., small remains small; Caillaud and Jullien 2003). One catalyst for growth that falls under the category of “growth hacking” is for the platform to attract and onboard marquee sellers. Marquee sellers are typically sellers who have a prestigious brand value and can bring recognizable supply to the platform. Their presence on the platform and associated name recognition attracts new buyers and often succeeds in kick-starting a positive cycle of growth. For example, one year after the release of Xbox (a gaming console), Microsoft acquired Bungie, a well-known game developer. Microsoft made this acquisition so it could make Bungie's popular game Halo available on Xbox, which was done in an effort to expedite the expansion of Xbox's user base (Edelman 2015). Another example is Huff Post, a leading media platform, who launched its business by hiring well-recognized writers to create popular blog posts to attract readers (Choudary 2016).

Although Microsoft's Bungie acquisition and other similar examples succeeded, not all marquee-seller growth strategies have been as fortunate. For example, Amazon Marketplace invited large sellers from China to list their products on its site. Unfortunately, it emerged that some of these sellers were selling counterfeit products and hence developed a poor reputation with associated negative externalities on Amazon's marketplace. As a result of the entry of the disreputable sellers, the supply from reputable incumbent sellers eventually plummeted (Shepard 2017). This failed example highlights an often-overlooked element in the cycle of growth strategy: the reputation of a marquee seller and its impact on existing sellers and the overall ecosystem of the platform.

1.1. Research Setting and Methods

We conduct an empirical study coupled with a stylized theoretical model to highlight underlying mechanisms for our empirical findings. To execute our empirical study, we partner with a leading online business-to-business (B2B) platform where retailers liquidate their excess/returned inventory. Online liquidation platforms, like the ones we study are growing and amounted to more than 600 billion USD of business in 2020 (Schoolov 2022). The merchandise on these platforms—including electronic items—are a mix of unsold and returned products. Hence, these B2B marketplaces serve a crucial role in extending the life of products, thus creating significant economical and environmental value.

Within its online community, our partner platform operates several distinct private marketplaces (PR marketplace) for individual, well-known marquee retailers (e.g., Walmart, The Home Depot). In addition, the platform operates a public marketplace (PU marketplace) where unbranded sellers (typically small- and medium-sized retailers) list their products. Having a healthy public marketplace is often of strategic importance for many of these online platforms as it allows them to have greater visibility into buyer and seller behaviors and to obtain greater detailed and timely information regarding market dynamics and trends. The platform features categories such as Appliances, Consumer Electronics, and Apparel, oftentimes with individual marketplaces offering a variety of categories on their sites. All wares are sold via (proxy) English auctions.

Given the wide range and condition of products sold on the platform (e.g., some products are brand new inventory; others are product returns, etc.), buyers rely on sellers' product descriptions to assess the value of the products. Although the platform supports a fairly standardized template for all auctions hosted on its site (see Figures EC.12 and EC.13 in Online Appendix EC.3), individual sellers vary in the degree of information they provide. Furthermore, some sellers' descriptions of listed products differ significantly from their actual

states. Such discrepancies often result in buyers submitting disputes to the platform and seeking financial compensation from the seller and/or platform. Although in PR marketplaces, the branded seller's name provides a signal of trustworthiness of their listing information (and hence the likelihood that a dispute will arise), the anonymity of sellers on the PU marketplace poses an added layer of challenge for creating efficient markets. In an effort to bring greater transparency to and to improve the efficiency of the PU marketplace, the platform tracks these disputes and posts for each seller a historical dispute rate (akin to ratings on other platforms). This dispute rate serves as a useful measure of a PU seller's reputation, hitherto referred to as a seller's *quality*.¹

We exploit the first two major (single category focused) marquee entry events on the platform to investigate our specific research questions discussed below. Examining single category entry events allows us to identify treatment and control categories to causally estimate our effects. A key aspect of these two marquee entrants is their differing quality. Although the first entrant is a very high-quality seller (with observed disputes close to zero), the second marquee entrant suffers from a far higher dispute rate than the typical PU marketplace seller (i.e., the second marquee entrant is a very low-quality seller). We thus refer to the first entrant as a *superior entrant* (SPR) and the second as an *inferior entrant* (IPR). These entries were staggered over three years (presumably providing sufficient time for the PU marketplace to react and adjust to a new norm of platform dynamics). We use a difference-in-difference-in-differences method for causal identification of the marquee entry effect. We then use a stylized theory model to better understand the mechanisms of our empirical findings and to assess the boundaries of our findings.

1.2. Research Questions

We focus on studying the impact of marquee seller entries on the growth of the PU marketplace. It is clear that a marquee seller changes the supply on the platform, which in turn exerts competitive pressure on existing PU sellers. It is not immediately clear if PU sellers are necessarily worse-off as a result. A key feature of contemporary online platforms that informs the ensuing supply-demand dynamics and its net impact on PU sellers is their *supply flexibility*; online sellers may be able to alter their supply quantities more easily than their brick-and-mortar counterparts. This in turn has implications for the portfolio of supply-quality offerings on the platform.

In addition, a marquee entry may also change the size and nature of demand facing a PU seller. More specifically, an entrant *can* have a differential effect on the new demand a PU seller experiences, with high-quality entrants bringing quality-conscious buyers, and low-quality entrants bringing less quality-conscious (more

cost-conscious) buyers. The negative impact of entry for PU sellers must be considered against this possible (associated) demand growth.

In this online environment, we seek to answer the following two research questions:

1. Are high-quality and low-quality PU seller revenues differentially affected by the entry of a marquee seller, and, if so, does the direction of this difference vary with regard to the marquee entrant's *quality*?

2. Is the PU marketplace composition, that is, the (relative) number of listings by high- and low-quality sellers, impacted by the marquee entrant's *quality*?

1.3. Summary of Results

The existing literature has found that high-quality incumbent seller revenues increase relative to that of low-quality incumbent sellers after a marquee seller entry (Zervas et al. 2017, Reshef 2023). We find that this is true *only* if the entrant is of *superior* quality (SPR). When the entrant is of *inferior* quality (IPR), we find that low-quality PU seller revenues are relatively better. To understand this, we disentangle demand and supply dynamics. We find that the PU sellers lose some of their business to the marquee entrants. The buyers who previously shopped with high-quality PU sellers are lost at a higher proportion than those who shopped with low-quality PU sellers when the SPR joins, whereas the opposite is true when the IPR joins. We term this as a *localized market-stealing effect*. Although some extant demand is lost to the entrants, PU sellers are able to attract new marquee-generated buyers. We find that high-quality PU sellers attract a higher proportion of these new buyers compared with low-quality PU sellers when the SPR joins, and that the opposite is true when the IPR joins. We term this as a *localized market expansion effect*.

Faced with these competing demand expansion/contraction forces, we turn to the PU sellers' strategic supply responses. We find that, although high-quality PU sellers expand their supply share relative to low-quality PU sellers when the SPR joins, the low-quality PU sellers disproportionately expand their supply when the IPR joins. The net revenue impact in our competitive auction setting is relative revenue gains/losses aligned with localized market expansion effects.² An associated outcome of these supply responses is that the average quality of PU marketplace listings shifts in the direction of the entrant's quality. If platform managers pursuing expansion fail to consider this, an unanticipated shift toward one end of the quality spectrum might ensue in the PU marketplace.

Our theoretical model allows us to highlight the key drivers behind asymmetric PU marketplace outcomes based on entrant type: (i) quality-sensitive (quality-insensitive) buyers brought in by high-quality (low-quality) entrants and (ii) the PU seller's ability to adjust

their supply quantities. Devoid of these features, the effect of entry are consistent with extant norms of thought. Absent (i), the PU sellers of higher quality would fare similarly or better than lower-quality ones upon any marquee entry, and absent (ii), PU sellers' revenues for those neighboring in quality to marquee entrants would be relatively worse off.

This paper makes the following three novel contributions to the literature on online platforms. First, to the best of our knowledge, this is the first paper to examine the externality of the marquee entrant's *quality* (i.e., reputation) on PU sellers' revenues. We find that results from previous studies do not generalize to entrants of varying quality. Second, we identify PU seller supply responses, allowing us to predict shifts in PU marketplace quality after entry. This is quite important to highlight given the quantity competition nature (accompanied by price-taking behavior) of unbranded sellers on many current merchandise and service platforms. Third, our theoretical analysis sheds light on the mechanisms driving the empirical findings and allow us to draw boundaries where the observed empirical findings hold.

The rest of the paper is organized as follows. Section 2 summarizes the related literature. Section 3 describes this study's empirical setup and data set. Sections 4, 5, and 6 present the empirical analysis. Section 7 discusses the mechanisms of our findings using a stylized model. We discuss the managerial implications and conclude the paper in Section 8.

2. Literature Review

Our work relates to two streams of literature in the fields of operations management, economics, and related areas. The first stream studies the effects of a competitor entry in online marketplaces, and the second stream focuses on seller supply behavior in online marketplaces.

Economic literature has deeply explored firm entry and its market effects. Generally, incumbents see decreased revenues due to increased competition from new entrants. Yet, higher-quality incumbents, in terms of products or services, outperform their lower-quality peers (Tirole 1988). For platforms, the dynamics are more complex. Although competition among sellers intensifies, the influx of additional buyers from new entrants counterbalances this. The overall impact on existing sellers remains ambiguous. Research on firm entry in online platforms is still emerging. In the context of bike-sharing (Cao et al. 2021) suggests that increased competition may benefit incumbents. Also, quality differentiates outcomes: Higher-quality sellers typically outperform their lower-quality counterparts. For instance, Zervas et al. (2017) found that Airbnb's entry negatively impacted lower-quality hotels more than their superior peers. Similarly, Reshef (2023) noted that

high-quality restaurants fare better when new sellers join a platform. Notably, existing research has not assessed how new entrants' quality/reputation affects current sellers, a gap our paper seeks to fill.

When competitors enter, a key strategy in online marketplaces is adjusting supply quantity. Various studies, like Allon et al. (2023), Chen et al. (2022), and Hall et al. (2021), explore how incentives or fares impact driver hours on ride-hailing platforms. Cullen and Farronato (2021) notes sellers' supply elasticity concerning price, whereas Karacaoglu et al. (2018) focuses on location decisions amid competition. Musalem et al. (2023) and Kesavan et al. (2014) delve into the flexibility of supply's effects. Our contribution highlights the varying supply reactions of reputable and nonreputable sellers during marquee entries, emphasizing the role of entrant's reputation. This perspective unveils new insights into the shifts in the platform's average supply reputation.

Our results are driven by the indirect network effects of the heightened supply-side competition on buyers. Specifically, Reshef (2023) found increased demand for incumbents in food delivery due to more sellers. Bimpikis et al. (2020) observed that concentrated listings in B2B auctions on specific days increased buyer participation. Meanwhile, Cao et al. (2021) reported benefits for an incumbent bike-sharing platform when a new competitor entered. Adding to this literature, we find that a marquee entrant not only attracts new buyers to the platform but also determines the quality-sensitivity of attracted new buyers, subsequently impacting how existing sellers of particular qualities perform. This nuanced view on network effects, largely overlooked, is a focus of our paper.

This study, set within an online platform context, echoes findings from the brick-and-mortar retailing literature. Traditional retailers frequently colocate, suggesting advantages from competition (Ellison et al. 2010). However, many studies indicate that incumbent retailers face challenges when big-box retailers enter the scene (Gielens et al. 2008). Notably, the impact varies: Incumbents close in distance or similarity are more adversely affected (Vitorino 2012). In contrast, our research emphasizes that in online platforms, proximate incumbents might actually benefit from such entries. Broadly, our research contributes to the growing empirical operations management literature on B2B marketplaces, ride-hailing, and food delivery platforms (Cohen et al. 2022, Alhauli et al. 2023, Zhang et al. 2023). Methodologically, we align with studies like Jiang et al. (2021), which integrate empirical and theoretical approaches in operations management.

In sum, our paper offers three significant contributions. First, we delve into the influence of entrant quality, moving beyond the typical focus on single-quality competitor entries. Second, we analyze supply responses and the ensuing quality shift in a marketplace, broadening the previously researched revenue effects related to

marketplace expansion. Third, we highlight a nuanced network effect: the quality-sensitivity of incoming buyers, influenced by the quality of the entrant seller.

3. Background and Data

Our partner platform is a leading online B2B platform where retailers liquidate their excess/returned inventory using a proxy English auction mechanism. Our data are granular, and we observe all transactions and participants since the platform's start in 2010 through 2018.³ The platform operates a single *PU marketplace* and several *PR marketplaces* within the same ecosystem. Within the single PU marketplace on the platform, small- and medium-sized unbranded sellers list their products. These sellers' listings are anonymous (i.e., the buyer cannot determine the identity of the seller), with only the observable dispute rates providing seller quality signals to potential buyers. The PU marketplace spans several product categories. Some prominent categories are Consumer Electronics, Appliances, Jewelry, and Toys, which account for 49% of platform listings and 71% of total sales. Each PR marketplace hosts products from a single branded seller, such as Walmart, with the marketplace carrying the name of the seller.

3.1. Sellers

Each seller belongs to a single marketplace (PU or PR). Sellers list bundles of items for sale in auctions, with each auction listing containing several items of the same category-type (e.g., multiple units of iPhone 7 devices) and of the same condition (e.g., new, used, or salvaged), with its contents specified. Once a seller has been approved to sell on the platform, creating listings is seamless and requires no intervention from the platform. Each listing typically remains open for three days, at which time the item is either awarded to the highest bidder or is taken off of the site if no bids were submitted. About 89% of listed auctions are sold (i.e., receive at least one bid above their starting price). Of particular interest to this paper is the category of Consumer Electronics: More than 93% of Consumer Electronics auctions are successfully sold when first listed.

3.1.1. Seller Quality (Reputation). Buyers rely on sellers' product descriptions to assess the value of listings. However, not all descriptions are accurate, with some sellers displaying a higher rate of discrepancy in the described versus actual state of the products listed. Buyers who have won a listing and find such a discrepancy often notify the platform and seek its assistance as they pursue financial compensation from the seller. Table 1 lists the sources of disputes and their relative frequency for the Consumer Electronics category of the PU marketplace; we note that 70% of the disputes arise due to inaccuracies in the product description (i.e., not as

Table 1. Dispute Reasons in the PU Marketplace

Dispute reason	Percentage (%)	Related to product description?
Not as described	51	Yes
Shipping problem	21	No
Item shortage	12	Yes
Other	12	Unknown
Product not available	2	Yes
Item damage	2	Yes

described, item shortage, item damage). To bring greater transparency and improved efficiency to the PU marketplace, the platform tracks and displays each (anonymous) seller's dispute rate within each auction that a PU seller lists (see Figure EC.12 in Online Appendix EC.3 as an example). The dispute rate captures the fraction of a seller's listings in which the buyer disputes the transaction.⁴ Hence, this dispute rate can be viewed as a measure of the seller's quality (reputation). To simplify our discussions, we denote a PU seller as *high-quality* (*low-quality*) if their dispute rate is below (above) the median value of 16%.⁵

It is important to note that the dispute rate of sellers within a category is not correlated with the type (e.g., cell phone, printers, etc.) or the condition of the products (e.g., new, used, salvaged) listed by the seller; see Online Appendix EC.2.1 for supporting analysis. We also note that sellers typically list products of varying conditions (new, used, salvaged). In fact, only 4% of the sellers list products in a single condition (Online Appendix EC.2.4 provides supporting analysis). Thus, sellers in our setting are separable along the reputation dimension but not along the product condition dimension.

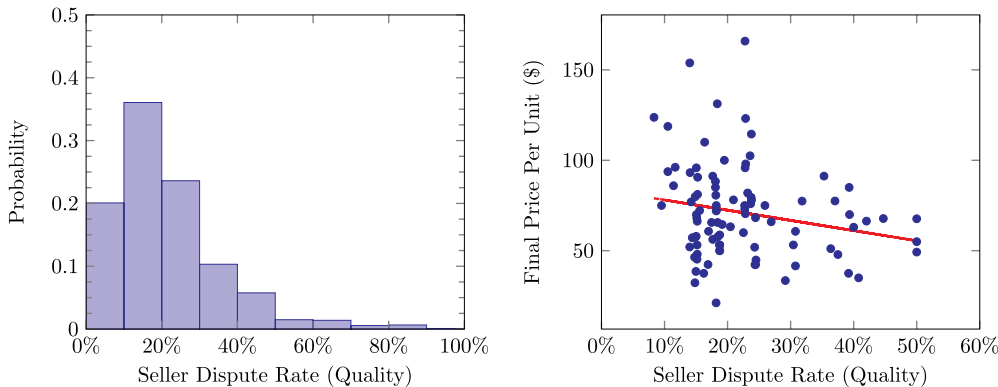
Figure 1 (left panel) displays the frequency and diversity of seller's quality (dispute rate) in the PU marketplace. These numbers highlight that disputes are not uncommon and that sellers' qualities vary widely across

the marketplace. Furthermore, the market demand rewards higher quality with higher prices. The right panel of the figure depicts the relationship between seller quality and final auction price for used iPhone 6 listings in 2016. The negative correlation suggests that, on average, buyers do respond to the quality information and bid accordingly, that is, buyers are *quality sensitive*. We also note that an individual buyer's experience with a seller is limited (a buyer bids in an average of 6 auctions throughout their lifetime, or 0.5 auctions per week). Thus, a seller's dispute rate should be an important consideration for buyers.

We also note that, in our data, an individual PU seller's quality (dispute rate) does not change significantly over time (i.e., high-quality (low-quality) PU sellers remain high-quality (low-quality) over time). This stability in dispute rates is reasonable in our setting given that sellers' item descriptions are a function of their downstream supply chain processes. Some sellers have tight control and visibility over the products that flow in their respective supply chains and are therefore able to describe their listings accurately and maintain low dispute rates, whereas others source from aggregators and resellers who do not pass along accurate information. This, in turn, leads to auction listings with poor or inaccurate descriptions, resulting in high dispute rates for these sellers. While sellers could plausibly adjust their quality (Prince and Simon 2014), we observe no such quality adjustments at the individual seller level.⁶ Also, Dinerstein et al. (2018) is a recent example of a platform study using eBay data where seller quality is a fixed attribute.⁷

3.1.2. Entrant Sellers. We focus our analysis on the first two major single category marquee entry events on the platform. A key aspect of these two entry events is their differing quality. The first entry event involves two

Figure 1. (Color online) PU Seller Dispute Rates and Buyers' Quality-Price Relationship



Notes. PU seller dispute rates (for all PU sellers) (left). Buyers' quality-price relationship in purchasing used iPhone 6 in 2016 (right). The slope of the fitted regression line is negative at the 0.05 significance level. The quality-price relationship is presented only for iPhone 6 in listings of year 2016 so that the effect of sellers' quality on final prices of similar products is illustrated.

large branded retailers whose marketplaces on the platform launched in January 2013 and March 2013. Given the almost simultaneous timing of their entries, the fact that both these entrants (marketplaces) were of high-quality with near zero dispute rates and that both marketplaces sold products in only the Consumer Electronics category, we consider them a single entry for our analysis. We refer to this entry event as the creation of a *superior* PR marketplace (SPR) seller. The second entry is of a large, branded retailer who launched its PR marketplace in July of 2015 and, similarly, only sold products in the Consumer Electronics category. During its first year of launch, this marquee entrant had an unusually high dispute rate of 39%, which is a particularly poor-quality level compared with unbranded sellers in the PU marketplace. Therefore, we refer to this entrant as an *inferior* PR marketplace (IPR) seller.⁸

Before our focal entry events, several PR marketplaces were created. Yet, each of these earlier entrants sold products across numerous major categories on the platform. This multicategory presence prevents us from identifying a reasonable control group, crucial for our analysis. Hence, our study focuses on the first entrants who sold exclusively in the Consumer Electronics category.

Further, although SPR and IPR sellers were not the only new PR entrants who sold Consumer Electronics during the four years covered by our observation windows, they were by far the largest and the only single-category entrants. Combined SPR and IPR sales accounted for more than 82% of all Consumer Electronics listings on the platform during the observation period.⁹

3.2. Buyers

The buyers on the platform are typically small resellers. In a typical week, about 33% of active¹⁰ buyers place a bid. On average, active bidders bid in three auctions per week, win 0.8 auctions, and purchase \$1,935 worth of merchandise. Buyers are free to participate in any of the marketplaces in which they are registered (PU or PR marketplaces).¹¹

We label buyers based on the marketplace in which they first register to bid; PU (PR) marketplace buyers are those whose first registration is on the PU (PR) marketplace. Further subdividing, SPR (IPR) buyers are those whose first registration is on the SPR (IPR) marketplace. In Table 2, we report buyer- and marketplace-level summary statistics. During the time period after the SPR's entry, PU buyers bid in 2.61 PU listings and 1.33 SPR listings, whereas SPR buyers bid in 1.99 PU listings and 11.28 SPR listings. Similarly, after the IPR's entry, PU buyers bid in 8.82 PU listings and 1.59 IPR listings, whereas IPR buyers bid in 5.64 PU and 15.77 IPR listings. These statistics imply that (i) bidders bid at a disproportionately higher rate in the marketplace in

Table 2. Buyers' Summary Statistics in Consumer Electronics Category

Buyer variable	Superior entry		Inferior entry	
	PU	SPR	PU	IPR
Number of buyers	814	227	2,251	345
In PU listings				
Average participated auctions	2.61	1.99	8.82	5.64
Average purchased auctions	0.74	0.44	2.25	1.36
In PR listings				
Average participated auctions	1.33	11.28	1.59	15.77
Average purchased auctions	0.19	2.48	0.15	2.37

Notes. Cross-bidding-related descriptive statistics are in bold. Statistics shown here are for postentry periods only (total of 17 months), because PR listings only exist after entry.

which they first registered, and (ii) bidders do cross-bid (participate) across marketplaces. This would suggest that bidders do actively search for products across markets and participate in those aligned with their price and quality preferences.

4. Empirical Setup: Identification Strategy

We proceed by analyzing the effect of marquee entry on the revenues and supply levels of PU marketplace sellers. We first describe our identification strategy, which we then execute in the rest of the section. A desirable identification strategy would randomly (in terms of both time and identity) add marquee sellers to the platform and study their implications on the PU marketplace. Because the adoption of sellers is costly, such a strategy would be hard to execute with large-scale platforms. We adopt the next best approach, wherein we study entries that are plausibly exogenous to the dynamics in the PU marketplace. We add a note of caution that such a strategy may not generalize to other settings, and may be violated in our setting as well.

We exploit two entries of Consumer Electronics marquee sellers to the platform. Although decisions with regard to which marquee entrant, and hence which categories, to add to the platform are strategic actions taken by the platform and can be endogenously determined, the *exact* timing of entry is plausibly exogenous as far as the PU sellers and all buyers on the platform are concerned. Addition of marquee sellers to the platform is a contractual process that involves discussion and negotiations between the platform and the marquee seller. Once the legal and contractual negotiations are finalized, it can take several weeks before a new PR marketplace goes live online. During this time, the platform does not make any preannouncements (e.g., "coming soon") to notify existing PU sellers or buyers of the impending entrant's arrival. In addition, the entries do not appear to be driven by any upstream economy-level shocks, as the growth in revenues in the entrant

categories (Consumer Electronics, Smartphones) in the United States were stable around the time periods included in the analysis (see Figure EC.11 in Online Appendix EC.3). Further, all of the marketing activities run on the platform are independent of PR and PU buyer/seller qualities and thus do not interfere with our identification.

Despite the plausible exogeneity of marquee entry, a before-after analysis of the impact of entry may still not be an appropriate and unbiased form of analysis due to significant time trends, as well as (unobservable to us) platform-level changes (e.g., website upgrades) around the timing of marquee entries. If unaccounted for, these may introduce significant bias into our estimates. To address this, we use, from the same platform, a control group that shares the same pretreatment time trend with our treated group.

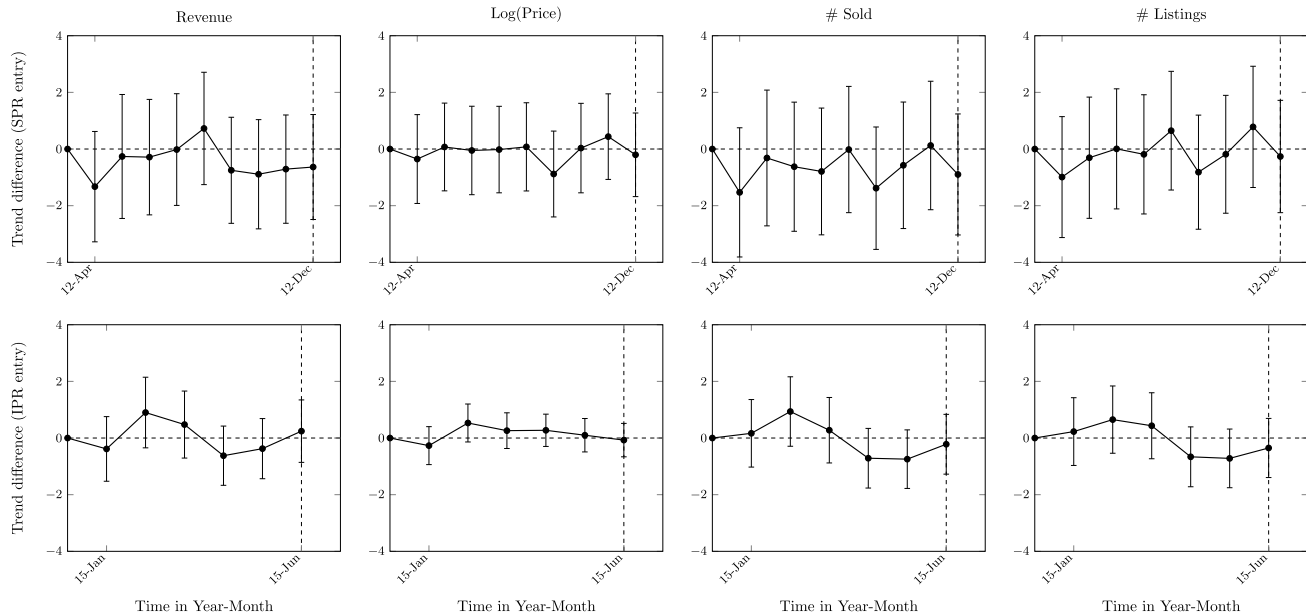
4.1. Control Group Selection

Both SPR and IPR entrants sold products only in the Consumer Electronics category. We treat this category as the *treated* group in each of the two main analysis time periods. We identify a control group (an alternate product category) for each of the marquee entrants such that (1) there is no PR entry in the control category in the analysis time period, (2) its PU sellers’ supply captures the counterfactual trend of the treatment group without marquee entries (e.g., PU marketplace’s organic growth), and (3) it is not affected by the entries of the selected Consumer Electronics marquee sellers. During the one-year analysis time-window of the SPR entry, of

all the categories on the platform, only the Jewelry and Toys and Unclassified categories did not experience a private market entry. Both these categories satisfy criterion (2) (shown by Figure 2 and Figure EC.1 in Online Appendix EC.1.1.5). Listings under the Unclassified category contain a mixed lot of various categories, including Consumer Electronics. Hence, the Unclassified supply might be susceptible to changes created by the SPR’s arrival and thus violates criterion (3). The SPR buyers bid in only 3% of the Jewelry and Toys listings, and the buyers of the Jewelry and Toys category bid in less than 0.1% of the SPR’s listings, satisfying criterion (3). For our analysis, we choose Jewelry and Toys as the control category. For the entry of the IPR, the only categories that satisfy criterion (1) are Appliances and Unclassified. Both these categories satisfy criterion (2). The PR buyers bid in only 0.1% of the Appliances listings, and the buyers of Appliances bid in only 0.1% of the IPR’s listings. Thus, this category satisfies criterion (3). We choose Appliances as the control category in this case. In addition, we affirm that the treatment and the control category listings share sufficient commonalities. In Table 3, we compare the major listing-level attributes (e.g., the retail values) between the treatment and the control groups. For all attributes, the difference of means normalized by the standard deviation is less than 0.25, suggesting adequate balance between the groups (Cochran and Rubin 1973, Ho et al. 2007).

We select the pre-entry and postentry analysis periods to exclude other private market entries/exits within treated and control categories. The SPR entries take

Figure 2. Parallel Trends: Trend Differences Between the Treatment Group and Control Group Before the Superior Entry (Top) and Inferior Entry (Bottom)



Notes. The vertical bands represent ± 1.96 times the standard error of each point estimate. The vertical dashed lines specify the end of the pre-entry period.

Table 3. Listing Comparison Between Treatment and Control Categories

Auction variable	Superior entry			Inferior entry		
	Treatment	Control	Difference/standard deviation	Treatment	Control	Difference/standard deviation
Retail price per auction (\$)	8,851 (6,182)	8,581 (13,005)	0.0207	12,500 (34,173)	11,959 (39,873)	0.0135
Retail price per unit (\$)	85 (65)	55 (153)	0.1992	343 (2,412)	407 (286)	0.0267
Units per auction	268 (520)	327 (276)	−0.1129	57 (47)	63 (56)	−0.1071
Probability of dispute per auction	0.1276 (0.1386)	0.1195 (0.1606)	0.0501	0.2115 (0.1899)	0.1646 (0.2452)	0.1914
Duration per auction (days)	4.0083 (1.4441)	3.8805 (1.4168)	0.0884	2.2791 (0.9814)	2.0253 (1.2420)	0.2043

Notes. Difference/standard deviation column shows the ratio of difference in corresponding treatment and control columns divided by the standard deviation of the difference. Standard deviations are noted in the parentheses.

place in January and March 2013. We select the pre-entry period starting from April 2012 to match the starting month of the postentry period. The postentry period extends until November 2013, as another marquee retailer who listed some Consumer Electronics auctions entered the following month.¹² The IPR entry takes place in July 2015. The pre-entry time window starts in January 2015, because a marquee retailer who specializes in Appliances launched its PR marketplace in December 2014, and the postentry period ends in May 2016, as a (relatively small) marquee retailer specializing in Consumer Electronics closed its PR marketplace the following month.¹³

4.2. Parallel Time Trend

To verify the parallel trend requirement between the treated group and the control group, we follow Autor (2003) by regressing the dependent variables of interest (i.e., PU sellers' revenues, average price, sold listings, and total listings) on the monthly dummy variables and an interaction term between the monthly dummy variables and the treatment group indicator, which gauges the trend differences between the treatment and the control groups. We also control for sellers' fixed effects.¹⁴ This check requires that the coefficients of the interaction term of the treatment indicator and month dummies for the pretreatment months are not statistically significant. Figure 2 displays the estimates of the interaction terms for the SPR and IPR entries. The vertical bands represent each coefficient's 95% confidence interval. As is evident in Figure 2, all pretreatment interaction term coefficients are not statistically significant. In other words, the parallel trend assumption holds between the treatment and the control groups.¹⁵

5. Seller Analysis

Our empirical specification examines the differences between the pre- and postmarquee entry period in the

treated group compared with the corresponding difference in the control group (treatment effect). In line with our research question, we focus on the difference between this treatment effect estimate for the high- versus low-quality PU sellers.

5.1. Model Specification

We specify a regression model to capture the effects of marquee entry on PU sellers of low and high qualities. The term $Y_{i,k,t}$ indicates four dependent variables for a PU seller i in category k and month t – PU seller i 's revenue, average normalized price (normalized price is the ratio of the sale price and the original retail price of a listing; we refer to this as *price* henceforth), number of sold listings, and number of listings (supply quantity). Trt_k is one if a category k is the treatment group and zero otherwise. $Post_t$ is one if a month t falls in a period after the focal marquee entry and zero if it falls in the pre-entry period. We use a binary variable $HighQuality_i$ to describe PU seller i 's quality, indicating whether their dispute rate is lower than the median (i.e., high-quality) or not (i.e., low-quality);¹⁶ $Trt_k \times Post_t$ captures the effect on low-quality PU sellers, whereas $HighQuality_i \times Trt_k \times Post_t$ captures the amount by which the entry effect on high-quality PU sellers exceeds that of low-quality PU sellers for the treated category. Then, $HighQuality_i \times Trt_k$ and $HighQuality_i \times Post_t$ controls for any quality-related differences in the treated group and postentry period, respectively. To control for any seller idiosyncratic characteristics (such as their location and other unobservable characteristics), we use seller fixed effects, η_i , whereas year-month fixed effects, $\zeta_{t\tau}$, control for platform-level monthly variations. All our dependent variables take positive values. As values for revenue, sale quantity, and supply quantity include a significant number of zeros (capturing months where seller i did not list any auctions), we employ a negative binomial regression for them.¹⁷ For price, which does

not take zero values, we use its log transformation as the dependent variable. The regression model is specified as follows:¹⁸

$$Y_{i,k,t} = f(\beta_0^{(1)} + \beta_1^{(1)} \text{HighQuality}_i + (\beta_2^{(1)} + \beta_3^{(1)} \text{HighQuality}_i) \times \text{Trt}_k + (\beta_4^{(1)} + \beta_5^{(1)} \text{HighQuality}_i) \times \text{Post}_t + (\beta_6^{(1)} + \beta_7^{(1)} \text{HighQuality}_i) \times \text{Trt}_k \times \text{Post}_t + \eta_i + \zeta_t), \quad (1)$$

where f is a negative binomial link function for revenue and number of listings regressions and an exponential link function for price regression.¹⁹

5.2. Estimation Results

Table 4 shows the estimated effects under the SPR and IPR entry. For the SPR entry, the insignificant coefficients on $\text{Trt} \times \text{Post}$ in columns 1, 3, and 4 show that the effect on low-quality PU sellers' revenues, sales, and supply is not statistically different from zero. In addition, column 2 shows that low-quality PU sellers' price decreases after the entry. In contrast, the coefficients on $\text{Trt} \times \text{Post} \times \text{HighQuality}$ indicate that high-quality PU sellers enjoy a positive effect on their revenues, prices, sales, and the number of listings relative to low-quality PU sellers. Interpreting the coefficients suggests that these effects are economically significant as well. The revenues, prices, number of sold listings, and number of all listings for a given high-quality PU seller per month show a relative increase of \$1,004, 3.0%, 2.28, and 2.53, respectively. Taken together, this implies that after SPR entry, the PU marketplace's supply as a whole expanded and that high-quality PU sellers benefited more from the expansion than their low-quality counterparts.

Turning to the IPR entry, the estimates of $\text{Trt} \times \text{Post}$'s coefficient suggest that the entry increases low-quality PU sellers' prices (column 2), whereas it decreases their revenues and sales (columns 1 and 3). In addition, PU sellers' supply reaction is not significant (column 4). However, the signs on the $\text{Trt} \times \text{Post} \times \text{HighQuality}$ are all opposite to those under the SPR entry. The high-quality PU sellers suffer more from the entry in terms of revenues, prices, sales, and supply compared with the low-quality ones (columns 1, 2, 3, and 4, respectively). The magnitude of these effects are substantial. The revenues, prices, number of sold listings, and number of all listings show a relative decrease of \$1,250, 5.9%, 1.61, and 2.31, respectively.²⁰ Taken as a whole, after IPR entry, the PU marketplace experienced shrinking revenues, whereas the high-quality sellers bore the brunt of this shrinkage.²¹ In Online Appendix EC.1.1, we conduct comprehensive robustness checks to strengthen the validity of the identified marquee entrants' heterogeneous effects on PU sellers.²²

In addition to the number of listings, we examine marquee entrant effects on other PU sellers' actions,

including entry, exit, size and item value per listing, category assortment across listings, and auction starting price. In contrast to PU sellers' salient supply reactions, none of these decisions show significant changes after each marquee entry; detailed analyses and discussions can be found in Online Appendices EC.2.5, EC.2.6, EC.2.7, and EC.2.8, respectively. Therefore, the estimated supply reactions of PU sellers tend to serve as their primary lever when coping with marquee entrants.²³

5.3. Discussion

Findings in Table 4 affirmatively answer both the research questions posed in the Introduction. We find that high-quality and low-quality PU seller revenues and supply quantities are differentially impacted and that the sign and magnitude of the effect critically depend on marquee entrant quality. In particular, our findings imply that, although the SPR's entry results in high-quality PU sellers being *better* off relative to their low-quality peers, the IPR's entry leaves high-quality PU sellers relatively *worse* off. A corollary of this analysis is the effect on the overall PU marketplace listing quality; given that the SPR versus IPR entry leads to differential supply responses by high- and low-quality PU sellers, we see that the average PU marketplace supply-quality shifts in the direction of the entrant quality. Based on the estimates, when the SPR joins, the fraction of high-quality listings in the PU marketplace increases from 69.1% to an expected value of 92.4%. In contrast, when the IPR joins, the fraction of high-quality listings on the PU marketplace decreases, on average, from 57.9% to an expected value of 31.7%.²⁴ Hence, the inclusion of a marquee seller on the platform not only affects the quality of the platform directly via the entrant's quality, but, as shown above, also has a substantial indirect effect via its effect on PU sellers. Marquee entrants exaggerate the shift of platform quality in the direction of their quality. If a platform does not incorporate this externality, it may set itself on an unanticipated cycle of growth skewed toward one end of the quality spectrum.

Our findings regarding the marquee entrants' heterogeneous effects on PU sellers naturally begs a deeper understanding of the mechanisms behind them. The results that both sold number of listings *and* prices move in the same direction for PU sellers after marquee entry cannot be explained if demand remains the same. In Online Appendix EC.2.9, we find that the number of average bidders per listing also moves in the same direction as supply and prices. This again points to changes on the demand side as a result of marquee entry. The following analysis looks closely at buyer behaviors, which, in turn, will guide us toward the mechanism supporting the effects described above.

Table 4. Heterogeneous Effects of Marquee Entrants on PU Sellers

Variable	(1) Revenue	(2) Log(price)	(3) No. sold listings	(4) No. all listings
Superior entry				
$Trt \times Post \times HighQuality$	1.575*** (0.441)	0.563*** (0.043)	2.042*** (0.511)	1.797*** (0.525)
$Trt \times Post$	0.052 (0.122)	−0.221*** (0.022)	−0.118 (0.123)	0.147 (0.130)
Observations	524	231	524	524
Inferior entry				
$Trt \times Post \times HighQuality$	−1.226*** (0.067)	−0.627*** (0.116)	−1.305*** (0.008)	−1.516*** (0.003)
$Trt \times Post$	−0.128*** (0.038)	0.478*** (0.111)	−0.085*** (0.00002)	−0.015 (0.013)
Observations	3,605	1,036	3,605	3,605
PU seller fixed effects	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes

Notes. Price is normalized and refers to the ratio of the sale price of a listing to its original retail price. It is missing if a given PU seller does not list any auctions. Therefore, the number of observations for the price regression (column 2) are fewer than those of the revenue and listing regressions. Clustered (category) standard errors in parentheses. The estimated effects of interest are highlighted in bold.

*** $p < 0.01$.

6. Buyer Analysis

In this section, we will show that buyers' behaviors (in particular, their quality sensitivity) are key drivers of the revenue and supply effects we observed above. Toward this end, we examine (sequentially) the participation and bidding behaviors of PU and PR buyers (recall that PU (PR) buyers are those who first registered in the PU (PR) marketplace). We will first show how PU buyers' participation changes in PU listings after PR entry and how PR buyers' participation in PU listings depend on PR entrant quality. These effects lead to localized market stealing and localized market expansion phenomena for PU sellers. In Sections 6.1 and 6.2, we elicit these effects based on a proxy buyer quality sensitivity measure. In Section 6.3, we precisely measure buyer quality sensitivity for a subset of data and validate our observations.

6.1. Localized Market Stealing

Localized market stealing occurs when the marquee entrant attracts PU buyers away from quality-neighboring PU sellers. To check for the presence of localized market stealing, we first examine the participation of PU buyers in PU and PR listings for the treated category. To examine the behavior through the lens of quality sensitivities, for each buyer j , we compute their (percentage) participation in high-quality PU sellers' listings before the focal entry, and denote it by $PctHighQualityParticipated_j$ (in log). This serves as a proxy for buyers' preferences for quality sellers (i.e., quality sensitivity). PU buyers who participate in relatively more high-quality PU seller listings are deemed to be *more* quality sensitive PU buyers, whereas those who participate less frequently in these types of listings are deemed to be *less* quality sensitive. Admittedly, this is an

imperfect proxy, and therefore we will return to capturing quality sensitivity with precision in Section 6.3. We note that PU buyers are consistent in their quality sensitivities between the before and after entry periods (see Table EC.34 in Online Appendix EC.2.10). A buyer with higher $PctHighQualityParticipated$ is more likely to participate in high-quality than low-quality PU listings in the postentry period.

The dependent variable $ParticipatePU_{j,t}$ is a dummy variable indicating if a PU buyer j participated in any PU listings in month t ; $Post_t$ is a dummy variable denoting the postentry period. The focal independent variable is $Post_t \times PctHighQualityParticipated_j$, capturing the difference in PU buyers' participation behaviors based on their preferences for higher-quality sellers. Using $PctHighQualityParticipated$ based on pre-entry time period requires us to leave out buyers who were not present in the pre-entry period. To control for buyers' idiosyncratic characteristics, we use buyer fixed effects, ν_j .²⁵ We also control for buyers' tenure (in months) on the platform.

To account for supply level changes that might differentially affect buyers based on their quality sensitivities, we apply supply level controls. To do so, the focal category (Consumer Electronics) is divided into three subcategories (cell phones, computers/laptops, and others, which includes tablets and gaming consoles); P_t denotes the number of PU listings in month t for each of these subcategories. We control for P_t and $P_t \times PctHighQualityParticipated_j$, which allows the control effect to vary with buyer quality sensitivity.²⁶ We also control for time effects using year-month fixed effects. The term $Z_{j,t}$ collectively denotes the control variables specified above (i.e., buyers' tenure, P_t and $P_t \times PctHighQualityParticipated_j$, year-month fixed

effects); we will reuse these in our subsequent buyer-level analyses. We note that our results are robust whether we control for $Z_{j,t}$.

The (logistic) regression model is given by

$$\begin{aligned} \text{logit}[Pr(\text{ParticipatePU}_{j,t} = 1)] \\ = \beta_0^{(2)} + \gamma_1^{(2)} \text{Post}_t \times \text{PctHighQualityParticipated}_j \\ + \beta_1^{(2)'} Z_{j,t} + v_j. \end{aligned} \quad (2)$$

To complement the above analysis, we examine PU buyers' participation in PR marketplace listings. Localized market stealing effects in the PU marketplace should imply relatively higher (lower) participation of quality-sensitive PU buyers in the SPR (IPR) marketplace. Verifying this also helps us rule out an alternative explanation that the suggested localized stealing effect on PU sellers is due to buyer attrition from the platform. Note that analyzing PU buyers' participation in the PR marketplace is only meaningful in the postentry period (i.e., the period after the entry takes place). $\text{ParticipatePR}_{j,t}$ is the dependent variable, which denotes whether a PU buyer j has participated in any of the new entrant listings in month t ; $\text{PctHighQualityParticipated}_j$ is the focal independent variable of interest. We use the same set of controls, $Z_{j,t}$, as before. We cannot control for buyer fixed effects anymore because they would absorb the effect of interest. Instead, x_j , which is defined as the average monthly number of auctions participated in by a buyer in the pre-entry period, is used to control for some buyer-level differences.²⁷ The (logistic) regression model for this analysis is given by

$$\begin{aligned} \text{logit}[Pr(\text{ParticipatePR}_{j,t} = 1)] \\ = \beta_0^{(3)} + \beta_1^{(3)} \text{PctHighQualityParticipated}_j + \beta_2^{(3)'} Z_{j,t} + x_j. \end{aligned} \quad (3)$$

Table 5 shows the estimates of Regressions (2) and (3) for both the SPR and IPR entries. For the SPR entry, the negative coefficient of $\text{Post}_t \times \text{PctHighQualityParticipated}_j$ in column 1 indicates that PU buyers with higher quality sensitivity decrease their participation in PU listings by a larger amount. Column 2 supports this; we find that PU buyers with higher quality sensitivity are more inclined to participate in SPR's listings. For the IPR entry, we see an opposite pattern: Column 3 indicates that PU buyers' participation decrease in PU listings is significantly less for PU buyers with high quality sensitivity. Column 4 further supports this by showing that these PU buyers are less inclined to participate in IPR's listings.²⁸

These findings align with the entrant drawing buyers away from PU sellers, increasing competitive pressure. However, how this pressure varies among PU sellers of different qualities is particularly intriguing. The entry of the SPR hurts high-quality PU sellers more by luring away quality sensitive PU buyers, whereas the entry of

Table 5. PU Buyers' Participation in PU and New Entrants' Listings

Variable	Superior entry		Inferior entry	
	(1) PU	(2) PR	(3) PU	(4) PR
$\text{Post} \times \text{PctHighQualityParticipated}$	-1.070** (0.501)		1.142*** (0.288)	
$\text{PctHighQualityParticipated}$		1.462* (0.873)		-1.212*** (0.390)
Avg. Participated Auctions (x)		-0.010 (0.053)		0.041*** (0.009)
Buyer fixed effects	Yes		Yes	
Year-month fixed effects	Yes	Yes	Yes	Yes
Supply assortment controls	Yes	Yes	Yes	Yes
Observations	13,780	7,638	69,788	28,229

Notes. Columns denoted by PU refer to ParticipatePU and by PR refer to ParticipatePR. Clustered (buyer) standard errors in parentheses. The estimated effects of interest are highlighted in bold.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

the IPR hurts low-quality PU sellers more by luring away less quality sensitive PU buyers who had predominantly shopped from them before. Because the competition effects of an entrant are particularly salient for PU sellers who are *neighbors* to the entrant in terms of quality, this confirms the existence of a *localized market stealing* effect.

This stealing effect stands in stark contrast to our observations in Section 5 that quality-neighboring PU sellers had a relative *boost* in revenue and supply quantity after an entrant joins. That is, the localized market stealing effect is unable to explain the observed revenue and supply outcomes for sellers of high and low qualities. We next examine the behaviors of PR buyers who join alongside the PR entrant, which helps us bridge the gap in our findings.

6.2. Localized Market Expansion

Localized market expansion is the relative increase in new demand of an entrant's quality-neighboring PU sellers. To check for its presence, we examine the participation of PR buyers in PU listings for the treated category and the postentry period by comparing PR buyers' participation in high-quality (H) and low-quality (L) PU sellers' listings. Unlike previous analyses where we distinguished buyer behaviors by $\text{PctHighQualityParticipated}_j$, here we distinguish the behaviors of PR buyers by SPR and IPR buyers.²⁹ We let the binary variable $\text{HighQualityListing}_q$ denote listings of high-quality PU sellers. It takes the value of one when $q = H$ and zero when $q = L$; Listing_q denotes the set of listings from PU sellers with the same quality (i.e., same $\text{HighQualityListing}_q$); and $\text{ParticipatePU}_{q,j,t}$ is our dependent variable, which indicates whether a PR buyer j participates in any listing in set Listing_q for $q \in \{H, L\}$ in month t . We use buyer fixed effects v_j and the same set of controls as in Section 6.1, including P_t , buyers' tenure, and year-month fixed

effects. The controls are collectively denoted by $\tilde{Z}_{j,t}$. The regression model is specified as

$$\text{logit}[\text{Pr}(\text{ParticipatePU}_{q,j,t} = 1)] = \beta_0^{(4)} + \beta_1^{(4)} \text{HighQualityListing}_q + \beta_2^{(4)'} \tilde{Z}_{j,t} + v_j. \quad (4)$$

A positive coefficient for $\text{HighQualityListing}_q$ would indicate that PR buyers disproportionately participate in high-quality PU sellers' listings. Table 6 reports the regression results for the SPR and IPR entries, respectively. We find that the coefficient on $\text{HighQualityListing}_q$ is statistically significant and opposite for the two entries. The positive (negative) coefficient for the SPR (IPR) entry suggests that the SPR (IPR) buyers participate relatively more (less) in high-quality PU marketplace sellers' listings. We note that the results are robust when we model buyers' participation for individual sellers rather than for a set of high- or low-quality sellers (see Online Appendix EC.1.3.1). In other words, PR buyers participate more in the listings of *quality-neighboring* PU sellers to the entrant, which results in *localized market expansion*.

The localized nature of market expansion suggests that buyers brought in by an entrant do have differing quality sensitivity and that the sensitivity is correlated with the quality of the entrant. Those brought in by the SPR entrant appear to be more quality sensitive and thus prefer to shop with high-quality PU sellers. This is despite high-quality listings having higher prices (33% higher, on average). The opposite is true for buyers brought in by the IPR entrant.

The localized market expansion effect is in the direction of the revenue and supply effect we saw in Section 5: PU sellers *neighboring* in quality to the PR entrant experience a relative boost in revenue and supply quantity after the marquee's entry. In Section 7, Proposition 1 will highlight that this *localized* nature of the market expansion is the key driver of the revenue and supply changes we observe. A *market expansion* that is not localized does not result in the asymmetric revenue and supply response that we have empirically observed.

Table 6. PR Buyer Participation in the PU Marketplace

Variable	(1) Superior entry	(2) Inferior entry
	<i>ParticipatePU</i>	
<i>HighQualityListing</i>	1.098*** (0.406)	-1.618*** (0.355)
Buyer fixed effects	Yes	Yes
Year-month fixed effects	Yes	Yes
Supply assortment controls	Yes	Yes
Observations	2,076	2,792

Notes. Clustered (buyer) standard errors in parentheses. The estimated effects of interest are highlighted in bold.

*** $p < 0.01$.

Given that from whom a buyer chooses to buy is a combination of their quality sensitivity and other factors (e.g., price and listing attributes), we take a closer look at buyers' behaviors to investigate the assertion that PR buyers' quality sensitivity is positively correlated with entrant quality. To do so, we focus our analysis on a subcategory for which there is a greater level of consistency and detail across all listings. For this subcategory, we examine buyers' bids (instead of just participation) in an auction and control for a more comprehensive set of listing features.

6.3. Buyers' Quality Sensitivity Comparison

Our focal platform and associated proprietary data set possess the advantage of having rich buyer-level information compared with a typical e-commerce platform. We are able to observe bidders' behaviors across a range of auction listings and marketplaces over time to estimate buyers' quality sensitivities. For identification, we use the variation within a buyer's bids across PU listings of different qualities. To prevent any listing characteristics other than quality influencing our quality sensitivity estimates, we focus on listings of iPhones where controls are relatively comprehensive because all iPhone listings state the model number (e.g., iPhone 4, 5, or 6), network carrier (e.g., AT&T), and condition information (e.g., new, used, or salvaged). We will therefore analyze buyers' bids for iPhone auctions and determine how the quality sensitivities of the PR buyers depends on the entrant quality and how these quality sensitivities compare with that of PU buyers.³⁰

Our dependent variable is a buyer's bid divided by the listing's retail value, denoted by $\text{Bid}_{i,j,\ell,t}$ (for PU seller i , buyer j , auction ℓ , and month t).³¹ We control for the iPhones' models, carriers, conditions, and per-unit retailer prices, as well as each listing's lot size (number of iPhone units), in vector $X_{i,\ell}$. The term IsPRBuyer_j indicates whether a buyer is a PR (= 1) or a PU buyer (= 0). The interaction term $\text{IsPRBuyer}_j \times X_{i,\ell}$ allows PR and PU buyers' to have different valuations for the product features. To control for any buyer-, seller-, and time-level unobserved factors, we add fixed effects for buyers (v_j), sellers (η_i), and year-month (ζ_t). The regression is specified as

$$\begin{aligned} \text{Bid}_{i,j,\ell,t} = & \beta_0^{(5)} + (\beta_1^{(5)} + \gamma_1^{(5)} \text{IsPRBuyer}_j) \times \text{HighQuality}_{\ell} \\ & + (\beta_2^{(5)} + \gamma_2^{(5)} \text{IsPRBuyer}_j)' \times X_{i,\ell} + \eta_i + v_j \\ & + \zeta_t + \epsilon_{i,j,\ell,t}. \end{aligned} \quad (5)$$

Table 7 presents the estimates.³² We find that PR buyers exhibit significantly different quality sensitivities than PU buyers. SPR buyers are significantly *more* quality sensitive than PU buyers ($\gamma_1^{(5)} = 0.019$), whereas IPR buyers are significantly *less* quality sensitive than PU buyers ($\gamma_1^{(5)} = -0.052$).³³ This analysis sheds light on

Table 7. Quality Sensitivity Comparisons Between PR Buyers and PU Buyers

Variable	(1) Superior entry	(2) Inferior entry
	<i>Bid</i>	
$IsPRBuyer \times HighQuality$	0.019* (0.011)	−0.052** (0.024)
Product and listing attributes	Yes	Yes
Buyer fixed effects	Yes	Yes
PU seller fixed effects	Yes	Yes
Year-month fixed effects	Yes	Yes
Observations	1,726	1,624

Notes. Standard errors in parentheses. The estimated effects of interest are highlighted in bold.

* $p < 0.1$; ** $p < 0.05$.

why the market expansion for PU sellers due to entrant is localized; entrants bring in like-minded quality buyers.³⁴

Although the extant marketplaces literature has shed light on the positive cycle of growth associated with supply side expansion, we find the effect to be more nuanced and intuitive. New marquee sellers on the platform attract buyers who have a *preference* for the quality type of these added sellers: High-quality entrants attract more quality-sensitive buyers, whereas low-quality entrants engage less quality-sensitive buyers. The localized market expansion and stealing competition that ensues determines the eventual (relative) winners and losers. Our results indicate that the positive or negative impacts of competition very much depend on whether the PU seller is neighboring (in quality) to the entrant. Notably, these buyer results remain robust across various tests presented in Online Appendices EC.1.2 and EC.1.3.³⁵

6.4. Discussion

Our empirical analysis has shed light on (i) the effect of PR entrant quality on revenues and supply of PU sellers (of high and low qualities), (ii) localized market stealing and expansion, and (iii) how localized market expansion is driven by the quality sensitivities of buyers brought in by the entrants.³⁶ Yet, a few questions remain, which the empirical analysis cannot help. First, why is there an asymmetric PU supply response based on PU sellers' quality? Second, would we continue to observe the same asymmetric quality-based revenue and supply effects if PR buyers' quality sensitivities were independent of the PR seller's type? Finally, are our revenue effects driven primarily by underlying changes in demand or by PU sellers' supply reactions?

Our empirical methodology of reduced form DiD is designed to causally estimate the new equilibrium that emerges after a marquee entry of a superior or an inferior quality. Although the DiD reduced-form approach

is an accepted approach to estimate equilibrium changes following a marketplace change (Li and Netesine 2020, Cao et al. 2021, Hall et al. 2021) and in general requires fewer identification assumptions than a structural modeling approach, we acknowledge that there are limitations. For example, a reduced form approach is generally not capable of allowing for rich counterfactual analysis to be conducted to answer questions such as those posed in the previous paragraph. To perform a counterfactual analysis given our reduced form approach, we turn to an appropriately defined theoretical model in the following section.

7. Mechanism of Empirical Findings: A Stylized Model

To identify the drivers of entrants' supply and demand effects and to address our remaining questions posed above, we propose a stylized game-theoretical model. Our goal is to use a simple model that can explain the localized market stealing and expansion effects of marquee entrant sellers on the PU marketplace. The model features (i) the presence of a PU marketplace that hosts sellers of different qualities and a pool of PU buyers with heterogeneous quality sensitivities; (ii) the entry of a PR seller with an exogenous quality, who brings along a pool of PR buyers to the platform; and (iii) sellers who optimize their revenues by competing in supply (i.e., number of listings) and buyers who choose the sellers to transact with so as to maximize their utility.

A key simplification we make in our theoretical model is that prices are formed by a Cournot competition rather than an auction model. The competing forces are similar across the two setups: sellers compete with supply, and prices are formed in the market. In fact, the literature (Bulow and Klemperer 1998, Klemperer 2004) has shown some equivalence between auction and oligopoly competition models. Using a Cournot competition model aligns with our goal of parsimony and vastly simplifies our analysis. Further, we only model two vertically differentiated PU sellers, one each of a high- and a low-quality. The market clearing prices under Cournot competition are such that all the supply is sold. This is, however, not too far from our empirical setup where we observe that almost all listings in our focal category (Consumer Electronics) are sold.

We describe the model setup in Section 7.1. In Online Appendix EC.4, we show that the model captures the empirical observations and intervening forces of localized market stealing and expansion. It also explains how these forces result in asymmetric PU supply response based on PU sellers' quality. In Section 7.2, we analyze the mechanisms that drive the empirical findings.

7.1. Model Setup

The model has two discrete time periods denoted by t ; $t = 0$ denotes a platform with only a PU marketplace

(buyers and sellers). At $t = 1$, a PR entrant joins the platform while also bringing in PR buyers. We model two PU sellers, one high-quality (θ_H) and one low-quality (θ_L), as well as one PR entrant (of quality θ_E). Each seller's marginal supply costs are normalized to zero; sellers decide their supply quantities (number of listings), $Q_{i,t}$. The resultant price(s), $p_{i,t}$, are a result of the Cournot competition. Sellers maximize their revenues $\pi_{i,t} = p_{i,t}Q_{i,t}$ in each time period.

There is a continuum of buyers, each of whose demand is infinitesimally small. The total mass of demand of PU buyers is one (normalized) and of PR buyers is μ . We focus on scenarios that assume PR buyers' mass is sizable but less than the PU buyers (i.e., $0.9 \leq \mu < 1$) to reflect our empirical setting (Table 2).³⁷ In line with our empirical findings in Tables 5 and 6, we model that buyers differ in their quality sensitivities, with some buyers willing to pay a premium price for higher quality. To capture this heterogeneity, we let x be uniformly distributed in the interval $[0, 1]$. The utility of a PU buyer x who purchases from seller i of quality θ_i at price $p_{i,t}$ is given by

$$U_{PU}(x, \theta_i, p_{i,t}) = (x\theta_i - p_{i,t}) \cdot \mathbb{1}\{\text{transact with seller } i\}, \quad (6)$$

where $x\theta_i$ stands for the buyer's willingness to pay (WTP) for seller i 's product. The utility of choosing the outside option is normalized to zero.

The utility specification for PR buyers allows them to have *different* quality sensitivities when compared with PU buyers. We define the utility for PR buyer y who purchases from seller i of quality θ_i at price $p_{i,t}$ to be given by

$$U_{PR}(y, \theta_i, p_{i,t}) = (y\theta_i^\alpha - p_{i,t}) \cdot \mathbb{1}\{\text{transact with seller } i\}. \quad (7)$$

Here, α captures the difference in PU and PR buyer quality sensitivities, whereas y , similar to x , is uniformly distributed in $[0, 1]$ to model heterogeneity among these buyers, and $y\theta_i^\alpha$ is the buyer's WTP for seller i 's product.³⁸ The utility of choosing the outside option is normalized to zero.

Note that for $\theta_i > 1$ for $i \in \{L, H, E\}$, $\alpha > 1$ ($\alpha < 1$) can capture that PR buyers are more (less) quality sensitive than PU buyers (i.e., $\theta_i^\alpha > \theta_i$ ($\theta_i^\alpha < \theta_i$)). Thus, without loss of generality, we assume $\theta_i > 1$ for $i \in \{L, H, E\}$ to allow α to capture quality sensitivity, and let α assume values within a unit-length interval centered at one (i.e., $\alpha \in [0.5, 1.5]$). In the subsequent analysis, we will analyze the entry of sellers who are of either much higher or much lower quality than PU sellers (i.e., a superior and inferior entrant) to reflect the nature of entrants in our empirical analysis. Consistent with the data, we assume that $\theta_E > \theta_H$ ($\theta_E < \theta_L$) for a SPR (IPR) entrant. As shown by Table 7, PR buyers' quality sensitivities are aligned with the entrants' qualities: PR buyers of the

SPR (IPR) entrant are *significantly* more (less) sensitive to quality than are PU buyers. To avoid α 's role be algebraically muted, we assume θ_i is sufficiently greater than 1 for $i \in \{L, H, E\}$ (note that $\theta_i^\alpha \rightarrow \theta_i$ when θ_i approaches one, muting the effect of α). Further, we assume that α is sufficiently larger than one (i.e., $\alpha \uparrow 1.5$) under the SPR entry, whereas α is sufficiently smaller than one (i.e., $\alpha \downarrow 0.5$) under the IPR entry.³⁹ Lastly, we exclude the cases where PU and PR buyers' quality preferences are widely separated (i.e., no overlap between PU and PR buyers' seller choices when θ_i is very large) by assuming that $\theta_i, i \in \{L, H, E\}$ are (sufficiently greater than one but) not too large. Online Appendix EC.5.1 provides a complete characterization of the assumptions. In Online Appendix EC.5.3, we numerically evaluate the robustness and boundaries of our analytical findings by relaxing our assumptions regarding the ranges of values that α, μ , and θ can take.

7.1.1. Market Equilibrium Outcomes. We focus on the pure-strategy Nash equilibrium of the supply-demand game before and after an entrant joins. The following set of equilibrium conditions hold: Each seller optimizes on their supply quantity, $Q_{i,t}$, given a price that is a function of their supply quantities, $Q_{i,t}$, as well as those of their competitor's total supply quantity, $Q_{-i,t}$. The optimal quantities satisfy

$$Q_{i,t}^* = \arg \max_{Q_{i,t}} Q_{i,t} \cdot p_{it}(Q_{i,t}, Q_{-i,t}^*).$$

Given each seller's quality and price, each buyer chooses a seller or the outside option (denoted by $\emptyset$) to maximize their utility. In particular, PU and PR buyers' optimal choices are given by

$$\arg \max_{i \in S_t} U_{PU}(x, \theta_i, p_{i,t}), \quad \forall t \in \{0, 1\}, \text{ and}$$

$$\arg \max_{i \in S_1} U_{PR}(y, \theta_i, p_{i,1}),$$

where $S_0 = \{L, H, \emptyset\}$ and $S_1 = \{L, H, E, \emptyset\}$. Let $\mathcal{X}_{i,t}^*$ and $\mathcal{Y}_{i,t}^*$ be the PU and PR buyer sets parameterized by x and y , whose optimal choice is seller i . That is,

$$\mathcal{X}_{i,t}^* = \{x | i = \arg \max_i U_{PU}(x, \theta_i, p_{i,t})\} \text{ and}$$

$$\mathcal{Y}_{i,t}^* = \{y | i = \arg \max_i U_{PR}(y, \theta_i, p_{i,t})\}. \quad (8)$$

Equilibrium prices are formed to match the seller supply and demand.⁴⁰ In other words, $p_{i,t}$ satisfies the following market-clearing conditions $\forall i \in \{L, H, E\}$:

$$Q_{i,t}^* = \lambda(\mathcal{X}_{i,t}^*) + \mathbb{1}\{t = 1\} \cdot \mu \cdot \lambda(\mathcal{Y}_{i,t}^*), \quad (9)$$

where $\lambda(\cdot)$ denotes the mass of a given buyer set.⁴¹

7.2. Model Analysis: Mechanisms for Empirical Findings

As a first step toward answering the questions at the end of Section 6.3, we verify with Observations 1 and 2 in Online Appendix EC.4 that our model captures critical empirical observations upon a marquee entry. These observations illustrate the model's ability to predict the effect of SPR and IPR entry on high- and low-quality PU sellers. These analyses suggest the following reasons for the asymmetric PU supply response based on PU sellers' quality. When faced with SPR's entry, there is an increase in quality-sensitive buyers (PR buyers). These buyers place a higher value from shopping with the θ_H than the θ_L seller. Increase of supply by the θ_H seller, and the resulting decrease in prices, helps θ_H in two ways: (1) it alleviates the market stealing effects of the entrant, and (2) it encourages PR buyers to shop with θ_H (localized market expansion). Because PR buyers associated with a SPR entry value quality relatively more than PU buyers, they are less responsive to decreases in prices of the θ_L seller; Hence, both market-stealing and expansion effects incentivize θ_H to expand supply relatively more than θ_L upon SPR entry. Despite a decrease in prices due to increased competition, the supply effect is stronger, and revenues for θ_H change in the direction of supply quantity. We find that the same set of mechanisms yield reversed results upon the IPR entry.

We now turn to a question posed at the end of the previous section; namely, would we continue to observe the same asymmetric revenue and supply effect on PU sellers if PR buyers' quality sensitivities were independent of the entrant's type? To study this, we compare our results above with a setting where $\alpha = 1$ for both the SPR and IPR entries. This is equivalent to assuming that PR buyers are similar (in valuation distribution) to PU buyers.

Proposition 1. *If $\alpha = 1$, the equilibrium ratio of supply from θ_H relative to θ_L , as well the equilibrium revenues, remain the same (are larger) after SPR (IPR) enters. Formally, $Q_{H,1}^*/Q_{L,1}^* = Q_{H,0}^*/Q_{L,0}^*$ and $\pi_{H,1}^*/\pi_{L,1}^* = \pi_{H,0}^*/\pi_{L,0}^*$ after an SPR entry, and $Q_{H,1}^*/Q_{L,1}^* > Q_{H,0}^*/Q_{L,0}^*$ and $\pi_{H,1}^*/\pi_{L,1}^* > \pi_{H,0}^*/\pi_{L,0}^*$ after an IPR entry.*

Proposition 1 states that the effect on PU sellers of an IPR entry is opposite to our empirical findings. There are two competing forces at play here, and the net effect is a result of which dominates. First, there is a quality effect— θ_H enjoys a relative quality advantage over θ_L . By expanding its (relative) supply and thereby reducing prices, θ_H attracts new buyers from both PR's and θ_L 's market (we note that the gain of some PR buyers is tempered by some loss of PU buyers to the SPR entrant). Second, there is a localized market expansion effect—when α is not restricted to one but reflects the quality of the entrant, the SPR (IPR) entrant brings in more high (low) quality-sensitive PR buyers. This, in turn, benefits

θ_H (θ_L) seller via localized market expansion. These two forces are in harmony under the SPR entry (both benefiting θ_H seller) while at odds under the IPR entry. When α is restricted to one, the localized market expansion effect is muted, which results in θ_H being relatively better off under both SPR and IPR entry. Devoid this restriction, under IPR entry, the localized market expansion effect dominates, which results in θ_L being relatively better off. Proposition 1 thus highlights the key role that PR buyers' quality sensitivities play in shaping the relative performance of θ_H and θ_L in the face of a new entrant.

As noted at the end of the previous section, it is worth disentangling the contribution of supply flexibility to the identified asymmetric revenue outcomes that flip based on the quality of the entrant. That is, although Observation 2 identifies revenue and supply outcomes when sellers are able to strategically adjust supply in response to a new entrant, we wish to isolate the importance of supply flexibility by characterizing equilibrium revenues postentry under the assumption that PU sellers *cannot* adjust supply levels in response to the entry.

Proposition 2. *Consider a PU marketplace where sellers cannot adjust their supply post PR-entry (denoted by NR; i.e., nonresponsive). The equilibrium ratio of revenues of θ_H relative to θ_L is smaller (larger) after a SPR (IPR) enters. Formally, $\pi_{H,1,NR}/\pi_{L,1,NR} < \pi_{H,0}^*/\pi_{L,0}^*$ after an SPR entry, and $\pi_{H,1,NR}/\pi_{L,1,NR} > \pi_{H,0}^*/\pi_{L,0}^*$ after an IPR entry.*

Proposition 2 states that in a market where PU sellers are unable to adjust supply in response to an entry, the quality-neighboring PU seller will be *worse off* than the quality-distant seller. One of the key drivers of our empirical findings was the presence of a localized market expansion which dominates the market stealing effects of an entrant. The ability to increase supply in the face of a new entrant can alleviate the market stealing effect and enhance the localized market expansion effect. Expanding supply lowers prices for the quality-neighboring PU seller and allows it to capture a larger share of the new demand. Proposition 2 highlights the critical role of supply flexibility in tilting the balance between market stealing and market expansion effects of the entrant on the quality-neighboring PU seller.

Proposition 2 also finds support in the empirical findings of Zervas et al. (2017) regarding the impact that Airbnb's entry had on hotel revenues. Airbnb's entry into the short-term lodging market is synonymous with a low-quality entrant in our setting, because it comprises a large number of economy accommodations. Similarly, hotels represent a PU seller with inflexible supply, for they cannot quickly change their room supplies in rapid response to Airbnb entries. Consistent with our proposition above, Zervas et al. (2017) find that quality-neighboring hotels to Airbnb (i.e., budget/economy)

are worse off than those of distant quality (i.e., luxury/upscale).

Our analyses above have explored the effect of an entrant with either superior or inferior quality compared with existing sellers. It is also reasonable to inquire into the effect of an entrant with intermediate quality (i.e., that falls in the middle range of quality offered in the PU marketplace)? We use our stylized model to explore this question under the assumption that the PR buyers' quality sensitivity mimics that of the (intermediate) PR seller. That is, when the entrant's quality is between θ_L and θ_H , the PU buyers' quality sensitivities should be similar to those of the existing buyers. We hence assume $\alpha = 1$ in this case (i.e., α is the same as that of PU buyers).

Proposition 3. *Consider a PR entrant of intermediate quality (i.e., $\theta_L < \theta_E < \theta_H$) and that PR buyers have the same quality sensitivities as PU buyers (i.e., $\alpha = 1$). The equilibrium ratio of supply from θ_H relative to θ_L , as well as the equilibrium revenues, are larger after the marquee seller enters. Formally, $Q_{H,1}^*/Q_{L,1}^* > Q_{H,0}^*/Q_{L,0}^*$ and $\pi_{H,1}^*/\pi_{L,1}^* > \pi_{H,0}^*/\pi_{L,0}^*$.*

Similar to Proposition 1, θ_H 's quality advantage over θ_L implies that it is better off relative to the latter after the intermediate-quality marquee seller entry. By (relatively) expanding its supply and thereby reducing prices, θ_H is able to subside its loss of PU buyers to the entrant and also cannibalize θ_L 's market. Proposition 3, along with Proposition 1, implies that regardless of the entrant's quality, high-quality PU sellers will never be worse off than their low-quality peers post-PR entry, as long as the demand quality sensitivity distribution does not change (i.e., $\alpha = 1$). In other words, it is not the direct effect of marquee entrants on the platform's supply (volume and quality), but their indirect effects on the buyers' quality sensitivities that determine the relative winners and losers in the PU marketplace.⁴²

8. Managerial Implications and Conclusions

Online marketplaces have a prominent role to play in the current economy. Liquidation B2B platforms like the ones we study are growing and amounted to more than \$600 billion of business in 2020 (Schoolov 2022). Growth via inclusion of marquee sellers is a strategy widely adopted by these and other early-stage online platforms to expand their offerings and accelerate the growth of their demand market. The implications of this strategy with regard to the welfare of existing sellers who differ in quality have not been previously studied and, hence, are not well understood. We bridge this gap in the literature by empirically studying the entry of two marquee sellers with very different qualities on (unbranded) existing sellers.

We find that, although high-quality PU sellers expand their supply share relative to low-quality PU sellers

when a superior-quality entrant joins, low-quality PU sellers disproportionately expand their supply relative to high-quality PU sellers when an inferior-quality entrant joins. This asymmetric supply response results in an asymmetric shift in the average listing quality in the PU market: The PU market quality shifts in the direction of the entrant. This externality is broken down into crucial mechanisms using empirical analyses and a stylized theoretical model. We highlight the contradicting influences of the market expansion and market stealing effects brought about by an entrant. We further delineate the role of the quality sensitivity of buyers brought to the platform due to the entrant and existing sellers' supply flexibility in moderating the entrant's competition effects.

Our study has direct implications for the growth of the liquidation platforms and online B2B marketplaces in general. One of the key managerial implications of our findings is that growth may produce both intended and unintended consequences for existing platform participants. Although growth should allegedly expand supply offerings to buyers, it may actually produce the opposite effect. Failure to incorporate the supply response of existing small and unbranded sellers may set a marketplace on an unanticipated cycle of growth skewed toward one end of the quality spectrum. A platform focused on maintaining a high-quality reputation for its public marketplace over the long-term might therefore want to steer away from inviting low-quality brands to the platform.

Similarly, unbranded sellers should approach which platform to join and build a reputation on with some caution. Somewhat counter-intuitively, joining a platform with a current and projected quality reputation (through marquee seller acquisition or otherwise) similar to theirs might yield benefits in the long run. In a similar vein, marquee sellers deliberating on what platforms to join might want to consider the quality of existing sellers. Joining a platform with seller qualities closer to theirs would give them access to buyers with a preference for their quality offerings.

For buyers considering which platform to join and build a know-how of, joining a platform with more sellers aligned with their quality preference could imply more choices as these sellers are incentivized to list larger number of items. Further, more similar quality sellers are likely to join in the future, given the upsides we have identified. There is a tradeoff, however, in terms of price. A platform aligned in terms of quality for buyers might also attract many similar quality-sensitive buyers and therefore require higher bids/prices to win an auction or procure a product.

The identified mechanisms also suggest actionable levers for platforms to moderate unintended consequences of entry. Platforms can target and attract quality-sensitive buyers via its marketing and customer

engagement activities. They can also influence flexibility of seller supply quantities by, for example, reducing frictions and making it easier for sellers to list and de-list on the platform, and using financing instruments to enable sellers to increase their quantities of listings.⁴³

It is important to note that our empirical setting possesses several key features that support our results. We reiterate these key features below and discuss the implications of our findings in related settings such as B2C marketplaces, peer-to-peer marketplaces, and gig-work platforms. Two key features of our setting are (i) price discovery through auctions and (ii) supply flexibility. Retail marketplaces such as eBay, ride-hailing marketplaces, and online labor marketplaces possess both these features and other similarities with our setting. For example, in some ride-hailing marketplaces such as Uber and Lyft, prices are set by the intermediary entity, although other ride-hailing marketplaces such as inDriver have taken the innovative price-discovery path and allow passengers to offer a fare to drivers based on their transportation needs and the driver's reputation. Drivers are matched with demand requests based on the submitted prices (bids) and current supply in the market. Further, as is noted in the literature (Hall et al. 2021, Chen et al. 2022, Allon et al. 2023), driver supply flexibility is a key feature of ride-hailing marketplaces. Online labor markets often engage in beauty contest auctions where freelancers are hired by clients based on their price bids, as well as their skills and past performances. Freelancers' hours and clients' use of freelancer hours can be viewed as flexible and responsive to the prices that emerge in the marketplace. Our theoretical analysis clearly underscores the importance of supply flexibility as a driver of our findings. From our model, we learned that the absence of supply flexibility favors high-quality suppliers. Our theoretical analysis, and its underlying Cournot model, also gave us insight into the role of price flexibility, that is, the ability of price to change as a result of shifts in market supply. Therefore, in marketplaces where price is set by the seller(s) and is responsive to competition changes, our results might continue to hold. However, in market settings with price stickiness (because of menu costs or other market frictions), it is reasonable to conjecture that price discovery or fixed prices should lead to different implications of marquee entries of differing qualities.

Another key feature in our setting is the characterization of a seller's quality (reputation) and the quality sensitivity of buyers. In our market, sellers sell multiple products, and the products themselves may be of varying conditions. Sellers' reputation is determined via the feedback provided by buyers. Specifically, the seller's dispute rate provides a measure of the trustworthiness of the information provided by the seller, that is, if the product(s) are actually as they were described by the seller. We found a seller's reputation to be independent

of their products' condition. We expect our findings to have implications for other online markets where reputation is a key driver of customer demand, such as online retail, ride-hailing, online labor, or food delivery marketplaces. Furthermore, we conjecture that our results will continue to hold in settings where product/service grade and reputation are positively correlated, provided that customers are quality-sensitive and positively value higher product grade and/or reputation.

Online marketplaces are pivotal in today's economy. Our analysis offers a framework for studying the impact of marquee entrants' quality on existing platform members. Although this study focuses on short-term effects, long-term implications, which are often aligned with short-term results in online platforms (Dinerstein et al. 2018), merit further exploration. Our findings mainly apply to early-stage marketplaces, where each marquee seller's entry significantly impacts the landscape. In mature platforms, the effect of marquee entry may be minimal. Furthermore, although our study assumes seller reputation is clear to buyers, when it is not, platform growth strategies might vary, presenting another research avenue. Exploring marquee entry's broader effects on platform quality and revenue is crucial for a holistic understanding.

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Endnotes

¹ We note that other commonly used notions of quality in online platform studies include product or service quality. Unless explicitly mentioned, our reference to quality refers to reputation. This is similar to the notion of quality used in the literature (Hui et al. 2016).

² It is interesting to note that, despite high-quality PU sellers enjoying a quality advantage, the low-quality PU sellers still manage to expand their share of listings after the IPR entry.

³ See Tables 1, 2, and 3, as well as Table EC.36 in Online Appendix EC.3, for the summary statistics of related variables.

⁴ Dispute rates are calculated as the percentage of recent transactions that resulted in a dispute. Rates based on both the past 90 days, and 12 months are shown to users. We used the dispute rate based on the past 90 days for our analysis. The breakdown of dispute reasons as shown in Table 1 is not visible to buyers.

⁵ An alternative dispute rate cutoff for high quality we use is the 25th percentile (see Online Appendix EC.1.1.2). Other sellers'

measures displayed by the platform are the number of completed transactions, purchases from repeat buyers, and unresolved dispute rate.

⁶ In Online Appendix EC.2.2, we formally show that marquee entry has no statistically significant effect on individual PU sellers' quality.

⁷ PR marketplace sellers have a dispute rate resolution and tracking mechanism, but it is not publicly displayed on the platform. Nonetheless, buyers frequently review PR sellers on platforms like YouTube and [trustpilot.com](https://www.trustpilot.com), offering insights into the entrant quality.

⁸ Note that prices are determined by auctions rather than by sellers. This allows us to not be concerned that our findings are a result of prices set by entrants and not their reputation.

⁹ In Online Appendix EC.1.1.10, we extend our analysis to several marquee entrants and report consistent findings.

¹⁰ We assume that buyers are active between the first and the last times they bid.

¹¹ Registration requires an application along with a valid business license.

¹² This results in 61 treated PU sellers and 23 control PU sellers. On average, within the pre-entry and postentry time windows, the treated PU sellers are active for 7.01 months, and the control PU sellers are active for 7.73 months.

¹³ This results in 437 treated PU sellers and 96 control PU sellers. On average within the pre-entry and postentry time windows, the treated PU sellers are active for 11.35 months, and the control PU sellers are active for 11.79 months.

¹⁴ Specifically, the model for the parallel trend comparison can be specified as $Y_{i,k,t} = f(\eta_i + \zeta_t + Trt_k \times \delta_t)$, where Trt_k is the indicator of the treated category, η_i stands for seller fixed effects, ζ_t denotes the monthly fixed effects of the control category, and δ_t represents the trend differences between the treated and the control categories.

¹⁵ Posttreatment period trend comparisons are shown in Figure EC.14 of Online Appendix EC.3.

¹⁶ Although the platform updates sellers' dispute rates with every new transaction, seller quality categorization (high, low) almost never changes in our data. We therefore do not denote seller quality to be time dependent for the clarity of exposition. See Online Appendix EC.2.2 for the analysis to support this claim.

¹⁷ We discretize monthly revenues at the \$1,000 level to make it a count variable. We also confirmed that the model captures zeroes well and there is no need to incorporate an additional zero-inflating mechanism.

¹⁸ The terms $\beta_1^{(1)}$ and $\beta_4^{(1)}$ are not identified due to fixed effects.

¹⁹ Our findings remain robust when choosing Poisson link function in Regression (1). Yet, Poisson regression is not chosen as the main model because its equi-dispersion assumption does not hold in our data. See EC.1.1.1 for detailed discussion.

²⁰ The fact that both prices and number of sold listings change in the same direction indicates that demand factors drive the results we observe. Alternative explanations such as price change for other reasons can explain revenue changes, but not the effect on number of sold listings.

²¹ The analysis of both marquee entries' average effect on PU sellers' revenues, prices, sales, and supply is shown in Online Appendix EC.2.12. We also note that the estimates in Table 4 remain robust when the standard errors are clustered at a seller level.

²² First, these findings continue to hold when we modify Regression (1) by replacing dummy variable *HighQuality* with alternative quality measures such as the dispute rate itself, which is a percentage

metric (Online Appendix EC.1.1.2), adding additional controls for market-level dynamics (Online Appendix EC.1.1.3), and excluding PU sellers' fixed effects to include all PU sellers (Online Appendix EC.1.1.4). Second, these effects remain significant under various sample constructions for estimation, including using an alternative control category (Online Appendix EC.1.1.5), applying standard DiD estimation to Consumer Electronics PU sellers without any control category (Online Appendix EC.1.1.6), trimming pre-entry and postentry windows with the same months of the year to account for seasonal factors (Online Appendix EC.1.1.7), and employing a granular sample with biweekly observations (Online Appendix EC.1.1.8). Third, in Online Appendix EC.1.1.9, we implement multiple methods (e.g., heteroskedasticity and autocorrelation consistent estimation of the covariance matrix of Andrews (1991)) to verify the findings' robustness to potential serial correlation. Finally, we find consistent effects when analyzing the effect of marquee entries in non-Consumer Electronics categories on PU sellers (Online Appendix EC.1.1.10).

²³ It is possible that a longer observational time window might show an effect on the entry and exit rates of sellers.

²⁴ Note that these postentry period numbers are predictions based on a model that time fixed effects remain the same. In actuality, the time trends on the platform show a decreasing trend in fraction of high-quality listings, and we observe that the actual fraction of high-quality listings in the PU marketplace after SPR entry is 80.9%, and after IPR entry is 49.2%.

²⁵ These buyer fixed effects absorb the direct effect of *PctHighQualityParticipated_j*, and hence, *PctHighQualityParticipated_j* is not controlled for.

²⁶ We did not/could not use these controls in the analyses in Section 5 due to the presence of seller fixed effects there. The unit of analysis here is buyer $\times$ month compared to the seller $\times$ category $\times$ month used earlier.

²⁷ In Online Appendix EC.1.2.1, we verify that buyers' pre-entry monthly number of bids is sufficient to control for buyer-level differences. We re-estimate $\gamma_1^{(2)}$ in Regression (2) with this replacement and find similar results.

²⁸ In Online Appendix EC.1.2.2, we show that the findings are robust to classifying PU buyers discretely into high- and low-quality sensitivity types as opposed to the continuous measure used above.

²⁹ *PctHighQualityParticipated_j* was defined for PU buyers based on their pre-entry behaviors and therefore cannot be applied to PR buyers.

³⁰ In addition, better controls also improve the efficiency of our estimates. Using simulated data, we demonstrate, in Online Appendix EC.2.11, the importance of including detailed product information in identifying quality sensitivity.

³¹ When a buyer bids on the same listing multiple times, we use their last bid in the analysis.

³² Some of the coefficients are not estimated due to fixed effects (Intercept, *IsPRBuyer*, and *HighQuality_j*).

³³ The identified sensitivity differences are robust and highly statistically significant in many of the cases when considering alternative quality specifications for PU sellers (see Online Appendix EC.1.3.3). Moreover, we find consistent estimates when analyzing buyers' participation decisions rather than bid values (see Online Appendix EC.1.3.2).

³⁴ In Online Appendix EC.2.3, an alternative analysis shows that PU sellers' quality plays an essential role in determining buyer bid amount, whereas sellers' nonquality attributes exert no statistically significant effect on bid amounts.

³⁵ Online Appendix EC.1.2 includes robustness checks for the localized market stealing. The resulting estimates remain robust when replacing buyer fixed effects with alternative buyer-level control variables (Online Appendix EC.1.2.1) or using a binary classification for buyers (Online Appendix EC.1.2.2). Additionally, the localized market stealing effect continues to hold while estimated using buyers brought to the platform by incumbent private marketplaces (Online Appendix EC.1.2.3). Online Appendix EC.1.3 establishes the robustness of the localized market expansion and the quality sensitivity difference between PU and PR buyers. In particular, these results continue to hold when using buyers' seller choice as the dependent variable (Online Appendices EC.1.3.1 and EC.1.3.2). Moreover, the buyers' quality sensitivity comparison results are robust to alternative quality measures (Online Appendix EC.1.3.3). Finally, the localized market expansion effect remains valid when estimated based on PR buyers' participation in incumbent private marketplaces (Online Appendix EC.1.3.4).

³⁶ In Online Appendix EC.2.10, we verify that PU buyers' quality sensitivities remain similar before and after both marquee entries.

³⁷ Note that a value of μ that is too low does not allow for a sufficient entrant-created market expansion effect, which played a pivotal role in our empirical analyses explaining PU seller revenue and supply changes. In Online Appendix EC.5.3, we numerically study scenarios when μ takes on smaller values.

³⁸ The exact quality preference α and size μ of entrant bought buyers would depend on entrant specific exogenous parameters such as their brand popularity, marketing efforts, and so on: factors that are outside the scope of this paper. We therefore take α and μ as exogenously given.

³⁹ Note that a value of α that is too low or too high may lead to behaviors that are not representative of our empirical setting. In either of these cases, the PR buyers would minimally bid/purchase from PU marketplace sellers marginalizing the externality of the entrant on PU sellers that we intend to study. This drives our assumptions on α above.

⁴⁰ The market-clearing condition matches our empirical setting where almost all Consumer Electronics listings are sold.

⁴¹ We can show that the market-clearing prices are equivalent in a Cournot competition and simultaneous auctions for given supply quantities under the following conditions: (1) a continuum of buyers and each buyer's demand is infinitesimally small, (2) each auction's supply is infinitesimally small and can only satisfy one buyer's demand, (3) buyers can place bids in all auctions, and (4) each auction is sold to the buyer with the highest bid.

⁴² Online Appendix EC.5.4 provides a closer examination of how the total PU marketplace revenues would fare under a marquee entry.

⁴³ For example, see <https://stripe.com/capital/platforms>.

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